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HCI 584: Python Application Development in HCI
Project Spec
6/15/26

Project Spec: DinnersReady

Project Description

This app would help households put all of their groceries to use throughout the week by:

- Suggesting recipes from available ingredients
- Track expiration dates
- Apply logic for number of servings
- Display nutritional information

The app will pull in recipes for the ingredients on hand, and will prioritize those ingredients who are nearing their expiration date. As the ingredient list dwindles or new ingredients are added, suggested recipes will change to make sure food is not being wasted.

The user should be able to import their grocery list if shopping online, and eventually take a picture of their grocery receipt and adjust the available ingredients within their DinnersReady account.

V1 (CLI): Using the free tier of TheMealDB API, a user can interact with the app via the terminal to retrieve recipe suggestions and simple recipe instructions based on their ingredient input.

V2 (GUI): Using what was built in V1, I would then like to use Streamlit for a more user-friendly experience. It has a built-in calendar date picker that I would like to use for food purchase and expiration dates. It would also help me present the ingredient and recipe data in a more digestible way. I would also like to consider using Edamam to retrieve more information on specific ingredients (like general shelf-life), without having to provide that information myself, hard-coded in a dictionary. Edamam has a number of different APIs that could help with this app, but I also am aware that it could easily lead to feature creep.

Task Vignettes

1. Recipe suggestion by ingredient(s) input by user

Cindy is very busy with her and her family's schedule, and planning meals is one thing that she would like to spend less time and energy doing. She knows that she has four chicken

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breasts in the fridge that need to be cooked or frozen today, and she also has four tomatoes that need to be used in the next day or two before they need to be thrown out.

Cindy opens DinnersReady, and chooses the option to 'Suggest recipes by ingredient'. She inputs chicken breast (4) and tomatoes (4) and selects 'Generate recipes'. She is presented with five simple recipes that feature those specific ingredients, with minimal other ingredients needed to complete the recipes.

Once selecting a recipe, Cindy is given the simple instructions to make the meal.

Details and ideas for later:

- Split the ingredients into two separate recipes
- Option to add additional ingredients that are presented in the suggested recipes (Example: a recipe for garlic tomato basil chicken lists all of the ingredients as either on hand or missing, and a user can click to add basil as one of their items they have on hand already in their kitchen)

Cindy is so busy with her
and her family's schedule -
and dinner is always a stressor.
She wants to use up the chicken
in the fridge and tomatoes before
they go bad. She puts the ingredients
into DinnersReady and gets 5 simple
recipes to choose from.

DinnersReady	
Input Ingredients	
#1	chicken(4)
#2	tomato (4)
#3	-----
#4	-----

DinnersReady	
Recipes	
#1	Skillet tomato chicken
#2	Garlic tomato basil chicken
#3	Chicken tomato fried rice
↓ more	

DinnersReady	
Skillet tomato chicken	
Ⓢ	4 chicken breasts
Ⓢ	4 tomatoes
Ⓢ	2 garlic cloves

- ✓'s for ingredients provided to DinnersReady
- X's for missing ingredients

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2. CSV grocery list import and recipe expansion

Tom is a planner and likes to submit his grocery order online for his five person household and pick up his grocery order from the store every Monday after work. He would like to import his grocery list into DinnersReady and receive ideas for what meals he can prepare for himself and his family all week. He knows what types of food they like, but would like to try different ways of preparing them.

Tom would like to modify the ingredients at upload, removing some grocery line items and adding others manually. He would like to be able to select meals to make, and have those ingredients then be removed or reduced from his DinnersReady inventory so that future suggested recipes take into account the new number of ingredients.

Tom would also like to adjust the serving size on the recipes to reflect how many he would be cooking that recipe for.

Details and ideas for later:

- Take a picture of a grocery receipt and generate / update the inventory in DinnersReady based on 'reading' the text from that receipt

Tom likes to pick up his weekly grocery order, so he shops online + picks up on Monday. He drops his entire grocery list into DinnerReady and gets ideas on what meals he can prepare for himself and his family all week.

DinnerReady

Import Grocery List

upload or paste

CSV

upload

DinnerReady

upload all ingredients?

☒ _____

☒ _____

☒ _____

☒ _____

☒ _____

☒ _____

☒ _____

☒ _____

☒ _____

☒ _____

Plan

- Remove or
add specific
ingredients

DinnerReady

Spinach egg bites
(makes 12)

Pasta bake
(makes 8)

Chili Fresca
(makes 14)

Burger Bowl
(makes 4)

DinnerReady

Spinach Egg Bites

☒ Spinach (best by 6/14 ... 3 days)

☒ 12 eggs

☒ 1 cup mozz

☒ 2 red peppers

- "best by"
for fresh
ingredients

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3. Personal nutritionist for healthy meal planning and expiration dates

Sally is a personal nutritionist who uses DinnersReady to help her clients expand their meal options, while keeping their favorite healthy ingredients included. Being able to export shopping lists for her clients helps them order and plan meals ahead of time.

Sally would also like to be able to easily access macro nutrition information and have her customers be alerted to expiration dates of the fresh produce they are buying. Recipes can include suggested ingredient additions in order to modify or build upon their initial simple recipes.

Details and ideas for later:

- Plan by day within app: Monday meal 1, meal 2, meal 3, snack 1, snack 2
- Suggested recipes based on past preferences, and encouraging new ingredient inclusion to add nutritional variety

Sally is a personal nutritionist who uses DinnerReady to help her clients expand their meal horizons, while keeping their favorite ingredients involved. Inputting shopping lists helps clients plan meals ahead of time, provides nutrition info, and alerts them to expiration dates of fresh ingredients.

DinnerReady	
Plan - Groceries	
☐	_____
☐	_____
☐	_____
☐	_____
☐	_____

DinnerReady	
meal 1	_____
meal 2	_____
meal 3	_____
Snack 1	_____
Snack 2	_____

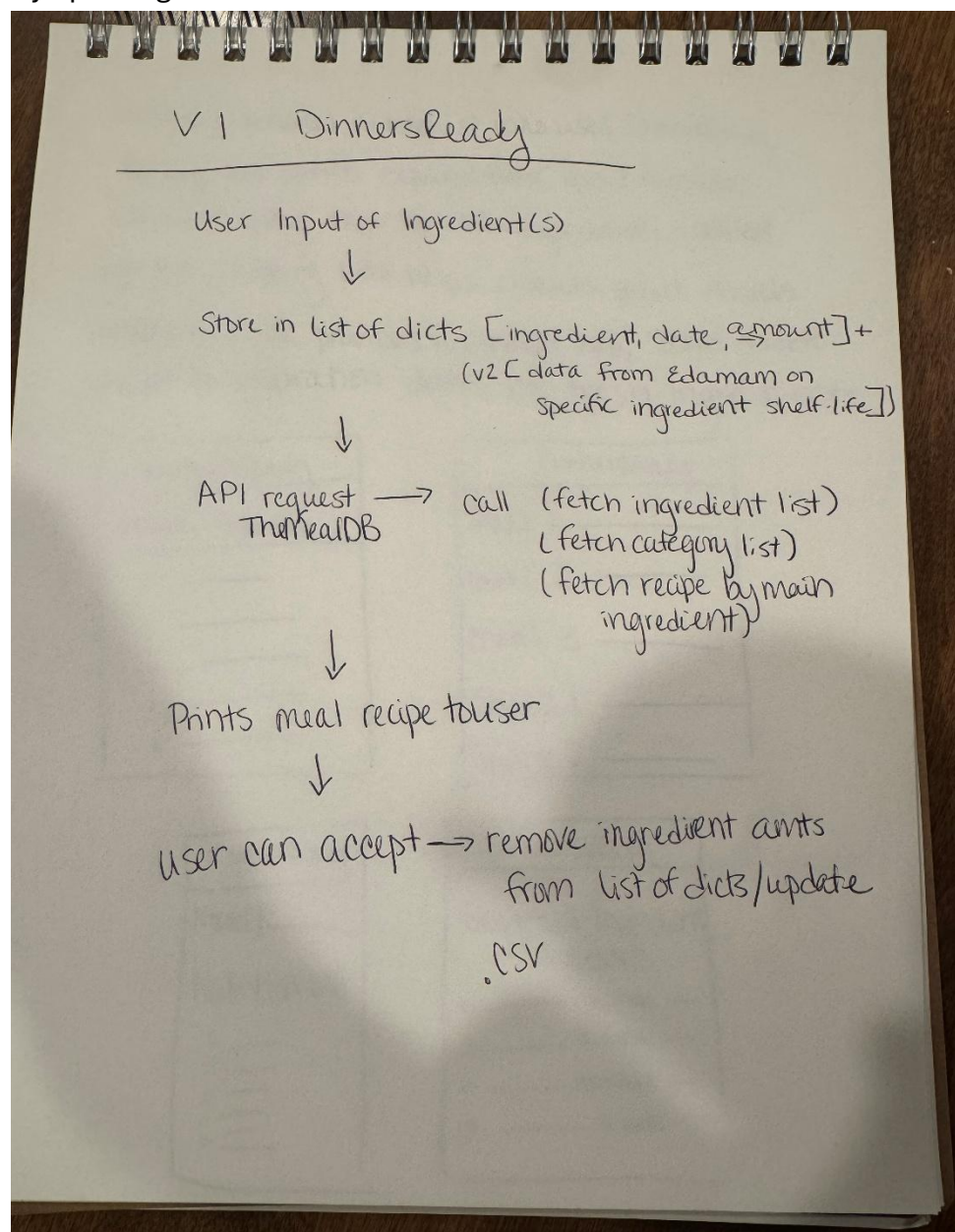
DinnerReady	
Meal 1 _____	
Nutrition Info	
☐	_____
☐	_____
☐	_____
☐	_____
☐	_____

DinnerReady	
Suggested Ingredient Additions	
☐	_____ to make _____
☐	_____ to make _____
☐	_____ to make _____
☐	_____ to make _____

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Technical Flow

I sketched out a bare bones version 1 of DinnersReady. It starts with user input, stores that information in a list of dictionaries, sends an API request to TheMealDB for information regarding that ingredient or category and a subsequent recipe suggestion. This suggestion would print in terminal to the user. If the user accepts this suggestion, the information stored in our dictionary needs to update and I'd like to store this information in an continuously updating csv file.



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MealDB is a free API with the universal key “1”. There are some premium methods, one of which I would like to use: Filter by Multi-Ingredient. I’m not sure what the proper etiquette is, but they also have a free method: List all Categories, Area, Ingredients. I looked at the List all Ingredients list (www.themealdb.com/api/json/v1/1/list.php?i=list) and wouldn’t I be able to loop through this list with multiple ingredients and pull out the recipes that have crossover? Or perhaps this is not proper and it is expected that I pay for the premium API.

One of the issues I had while I was testing out different search terms was that I typed in “pasta” to search in the ingredients list, and it returned ‘null.’ However, “pasta” is a category, which will display recipes with spaghetti, fettuccine, etc. I would like to loop through both the ingredient list and the category list to pull results from each, not display duplicates, without creating too much lag time.

I would like to integrate in V2 some of the functionality that Edamam has when it comes to shelf life and ingredient information. I would need help thinking through how and when to make calls to each app within the same workflow, or if I should be thinking about the workflow differently.

Final self assessment

The biggest change was looking into TheMealDB (your suggestion) and seeing that that API would be easy for a beginner like me to work with as it is free and has a public key of ‘1’ that made it easy for me to play around with.

One challenge is that I’d like to look up recipes by multiple ingredients which is a premium feature, but it isn’t too expensive and I may pay for it instead of trying to do the workaround by pulling multiple lists of data and then trying to cross reference and de-duplicate them.

Another challenge is that Edamam is talked about a lot, and it seems to have a lot of functionality, but it does overwhelm me a bit with all of their API options.

I would like to solve for the ingredient expiration considerations, and prioritizing those ingredients first. I would like to be able to do this without having to supply the logic myself and by using Edamam for that, but I am unsure how to implement a prioritizing logic to it to then get suggested recipes based on those ingredients from TheMealDB.

I am least familiar with working with different APIs and how to maximize what they offer for my own project, and when certain APIs should be handing off to my own engine I’ve built

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and pulling in information from another API. I would like your help getting my head around that part.

Original Sketch Idea

Sketch #3: No food wasted

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This app would help households put all of their groceries to use throughout the week into various recipes, while also tracking expiration dates and applying logic for number of people eating the meals. The app will pull in recipes for the ingredients on hand, and will prioritize those ingredients who are nearing their expiration date. As the ingredient list dwindles or new ingredients are added, suggested recipes will change to make sure all food is being put to use.

Users and Secondary Stakeholders

Primary users would be budget-conscious home cooks and busy families who need to relieve some of the mental load of trying to determine what's for breakfast and dinner each day. Secondary stakeholders could be personal nutritionists who are trying to help clients within their own budget and what's available to them.

The problem to be solved

This app could help prevent that bag of spinach or those grapes tucked away in the crisper drawer from going bad. Planning meals out ahead of time by reusing a lot of the same ingredients in different ways can help take the stress off of the person in charge of cooking, while knowing that they are also not letting anything go to waste. When new ingredients are acquired, new recipes become available.

The workflow

When returning from the store or receiving groceries from an online ordering service, the user can input those items into the app. As time passes, the user will be notified to high

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priority ingredients that need to be used. They can also select specific ingredients to find a minimum of 3 different recipes that incorporate them.

Data needed

User would input initial grocery inventory variables and calendar dates. The app would use the Edamam free food database api to help pull in ingredient information and suggested recipes.

Results

The user would input their ingredients and receive alerts on ingredients that need to be used immediately or have expired. The app would then suggest recipes to make with those ingredients. I would like to build this idea with a Flask app. I haven't used Bootstrap, but have seen that they can be quite handy for design templates. I would need help with using an external API to pull in the right type of requests.
